Online Shopping / E-Commerce System

Laravel (PHP) + MySQL — Project Overview and UML Design

Project Description

A Laravel web application backed by MySQL. Customers browse products, add items to a cart, check out, pay, track delivery, and review products. Sellers manage listings and fulfill orders; Admins manage users/listings; Couriers update shipment status.

Objectives

Convenient shopping; consistent cart/order/payment/shipment records; payment-gated order confirmation; seller fulfillment; courier tracking; administrative management.

Key Features

Customer: browse, cart, checkout, payment, track order, review. Seller: manage products and fulfill orders. Admin: manage users/listings. Courier: update shipment status.

Technology Stack

Laravel (PHP), MySQL, Blade, HTML/CSS/JavaScript, Bootstrap, Eloquent ORM, migrations, Git/GitHub, and PlantUML.

Setup / Execution

Install PHP, Composer, and MySQL. Create a Laravel app using `composer create-project laravel/laravel ecommerce-capstone`. Create a MySQL database and configure `.env` DB values, then run `php artisan key:generate`, `php artisan migrate`, and `php artisan serve`. These are setup instructions; application features still require implementation.

UML Standards

Class diagram uses class name/attribute/operation compartments, visibility, associations, and multiplicities. Sequence diagram uses lifelines, activation bars, call/return messages, and alternative fragments. Activity diagram uses swimlanes, initial/final nodes, actions, decision diamonds, guards, and control flows.

Required Scope

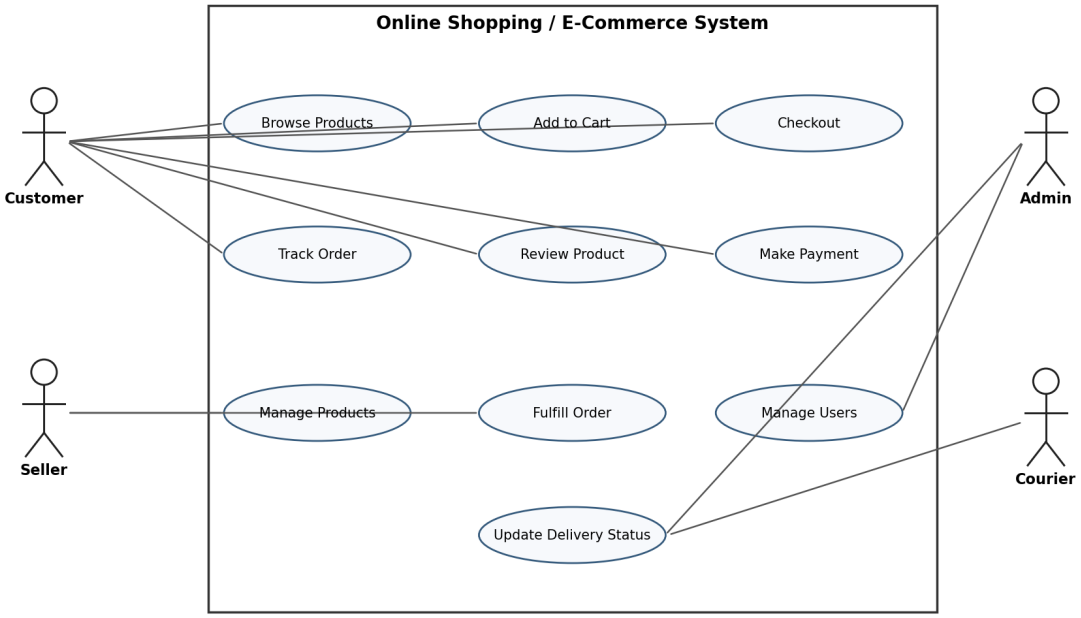
Actors: Customer, Seller, Admin, Courier. Sequence: Add to cart → Checkout → Payment → Order fulfillment → Delivery. Core classes: Product, Cart, Order, User, Payment, Shipment. No separate bank/payment-service actor.

UML Diagrams

Use Case Diagram

Required actors and assigned use cases.

Use Case Diagram — Online Shopping / E-Commerce System

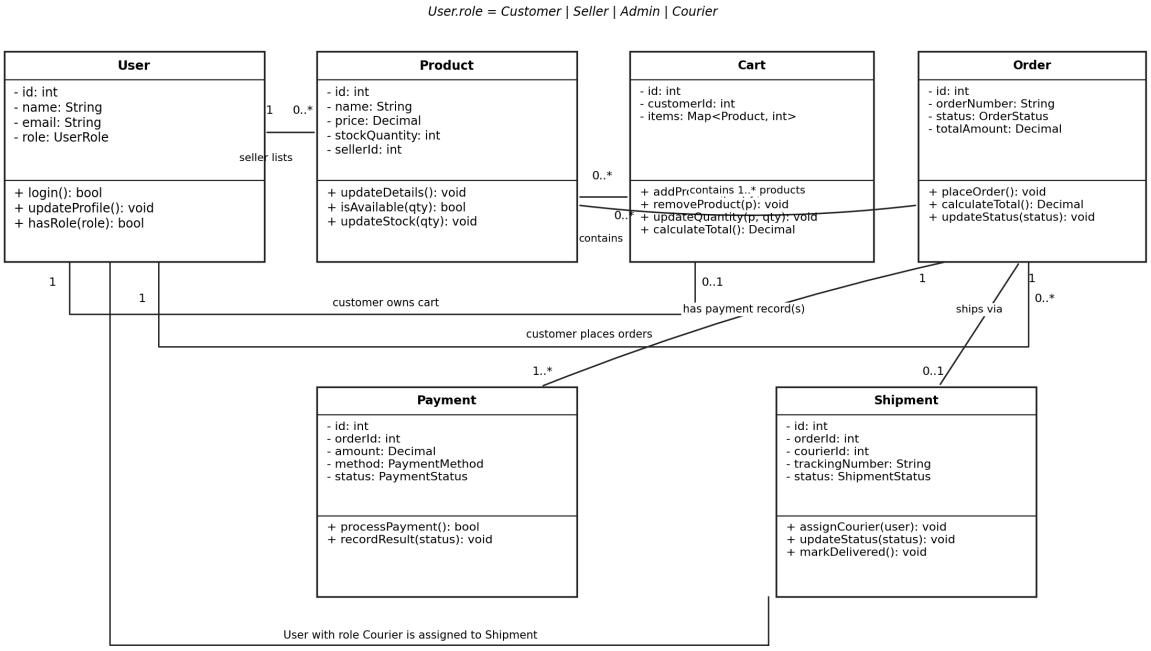


Payment is processed by the system; no separate bank/payment-service actor is modeled.

Class Diagram

Six required core classes with attributes, operations, associations, and multiplicities.

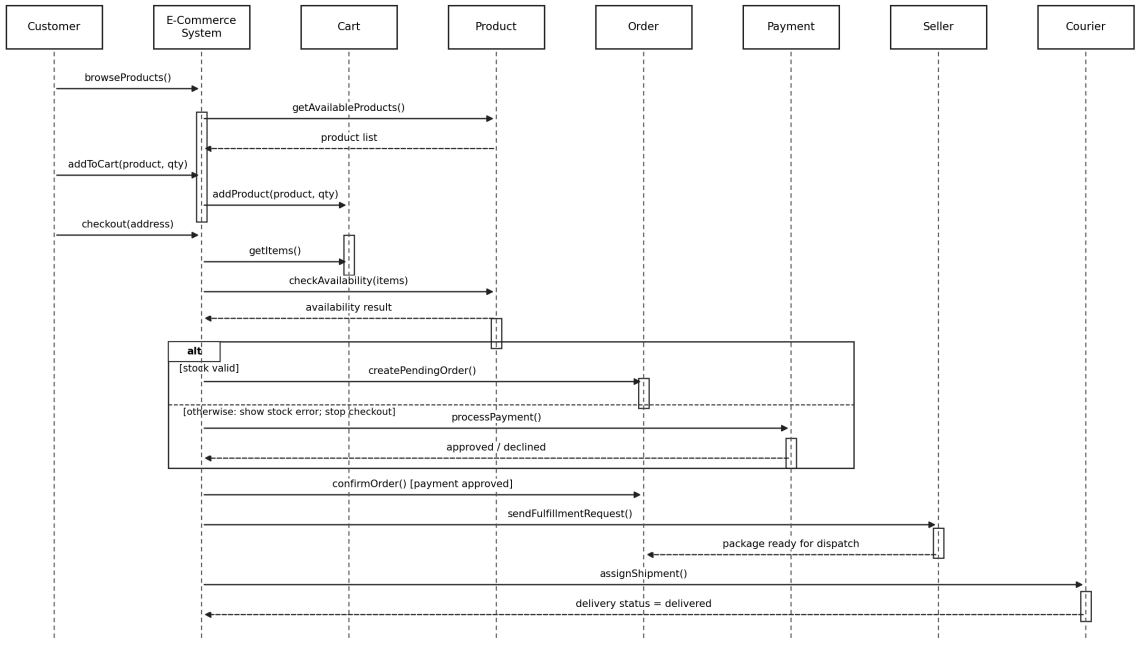
UML Class Diagram — Online Shopping / E-Commerce System



Sequence Diagram

Cart, checkout, payment, seller fulfillment, and courier delivery.

UML Sequence Diagram — Cart to Delivery



Payment declined: record failure, show retry/change-method option, and do not confirm or fulfill the order.

Activity Diagram

Swimlane workflow with stock/payment decision branches.

